Lesson for the course "Human-Computer Interaction"

HCI, IoT, and AI

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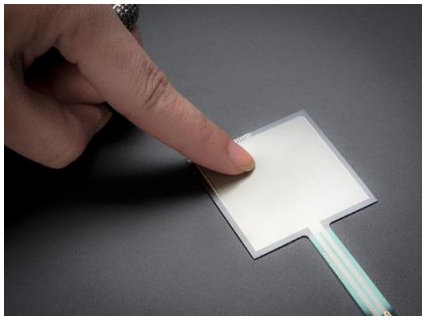


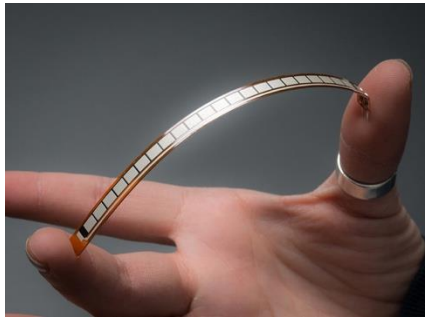
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Recap



- Smart objects
- Smart environments
- Services that they offer
  - Exam: these services need to be derived from user research









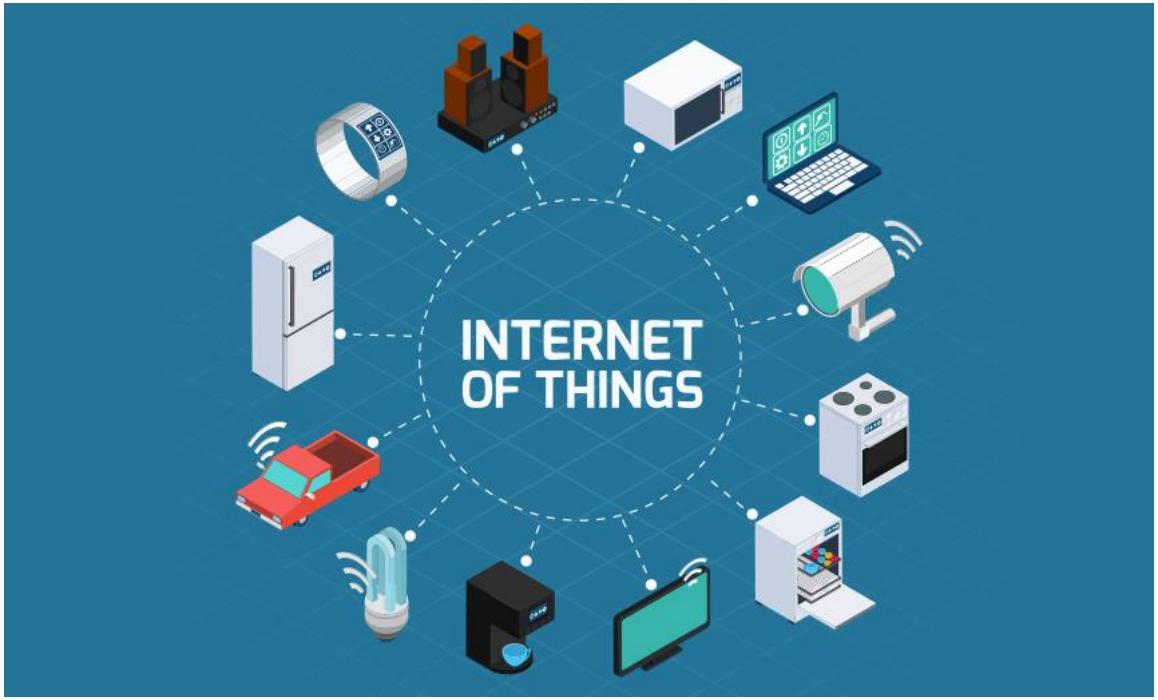
Recap

- Types of interactions
  - Discrete, continuous
  - Touch-based, touchless
  - Synchronous, asynchronous
  - Individual, collaborative
  - Situated, remote
  - Networked
  - Unimodal or multimodal
  - …and the combination thereof

What is the Internet of Things?

What is the Internet of Things?

- The Internet of Things (IoT) is the network of physical objects (or "things") embedded with ***electronics, software, sensors, actuators, and network connectivity***, which enables these objects to ***collect and exchange data***



What is the Internet of Things?

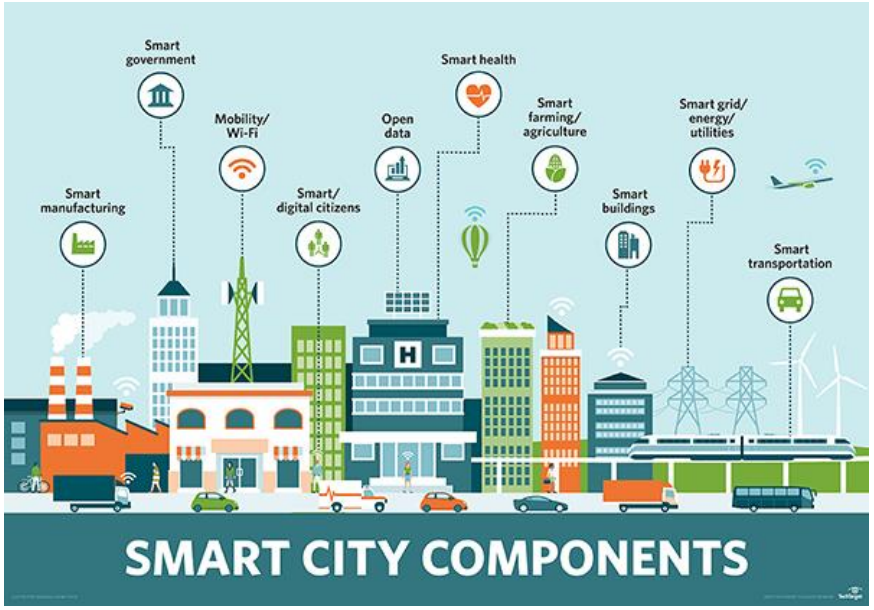
- **Things**
  - Are smart objects
  - Are physical objects augmented with sensors and computing technology, and connected to the internet
  - Capture data on the world around them and share them through online systems
  - Can be identified and tracked automatically
  - Have some form of **intelligence**



What is the Internet of Things?

- The Internet of Things **allows people and things to be connected anytime, anywhere, with anything and anyone**

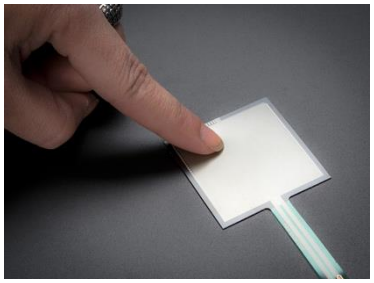




- Artificial Intelligence is increasingly used in IoT

Data fusion

- Data collected by sensors needs to be transformed into meaningful and task relevant information
- A system may have **various sources of information**, including multiple sensors and even data coming from social networks



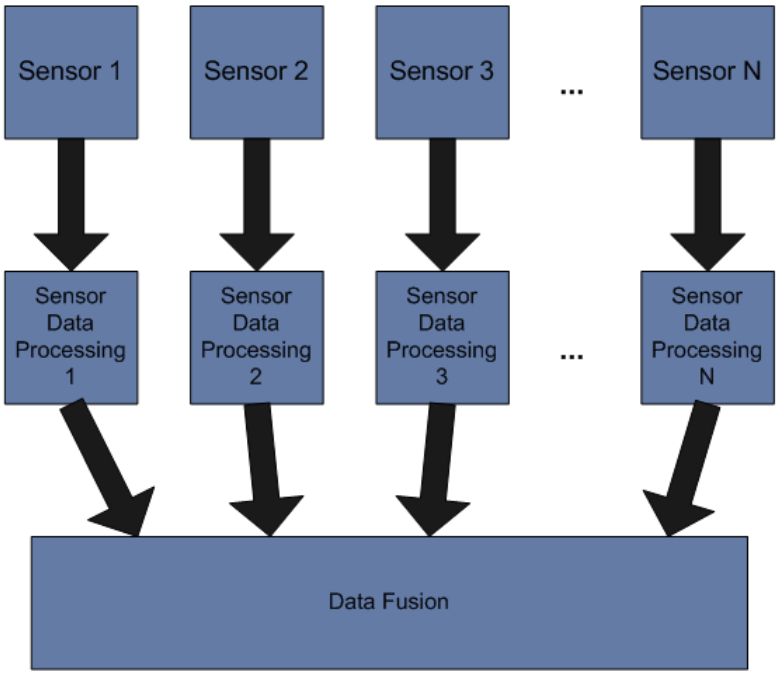




Data fusion

- Data fusion techniques combine data from multiple sensors and related information from associated databases to achieve improved accuracy and more specific inferences than could be achieved by the use of a single sensor alone

- The intelligence embedded in the system is responsible for **fusing data** and **extracting meaning** from it

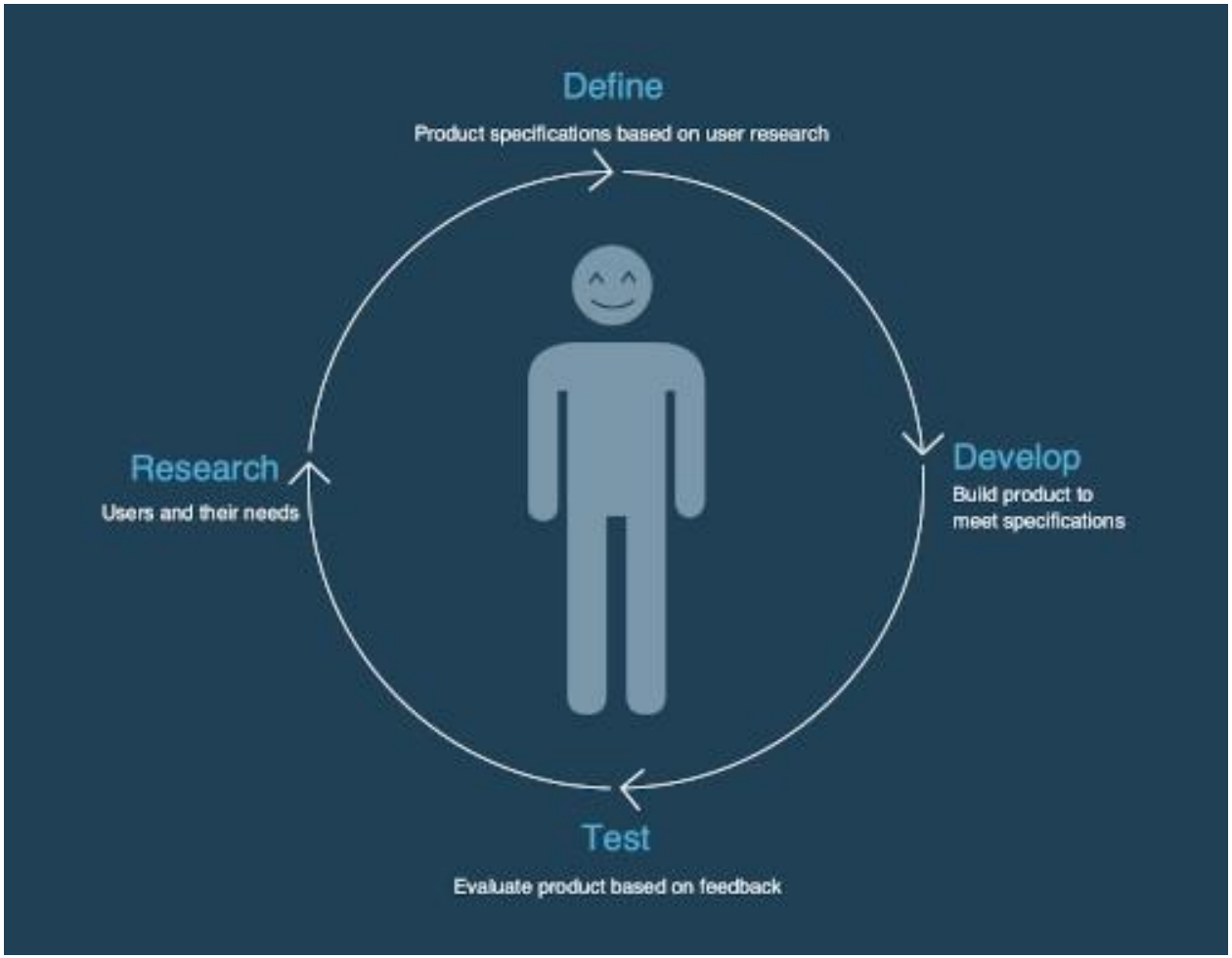


HCI and IoT

- HCI should focus on how computing could be added to everyday objects, and what this can enable for users
- **Combining data** from several smart objects/smart environments and interaction with them over time can result in **new knowledge** available to the user or the context aware environment, **to make suggestions for lifestyle improvements**
- Some goals for HCI in IoT
  - To create a useful link between the physical and digital worlds
  - To provide meaning from large amounts of data gathered
  - To enable users to learn about themselves, others and their surroundings
  - To support users in daily activities

How do you design for the people in the Internet of Things?

start

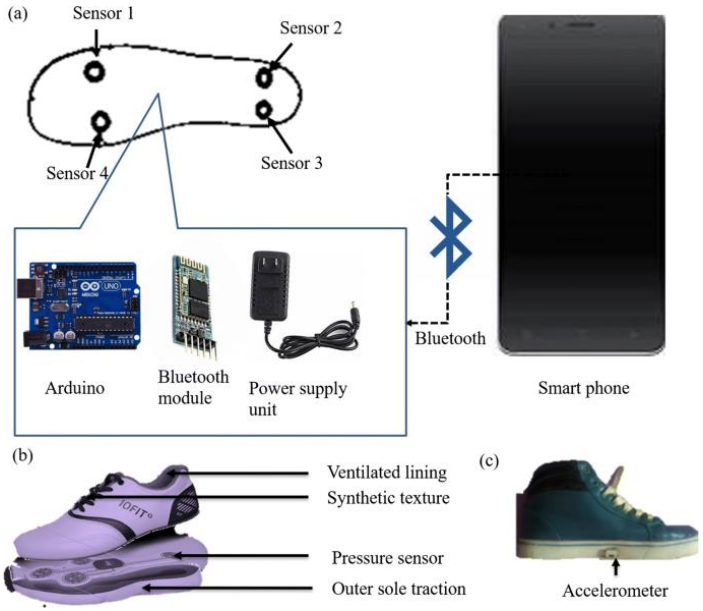


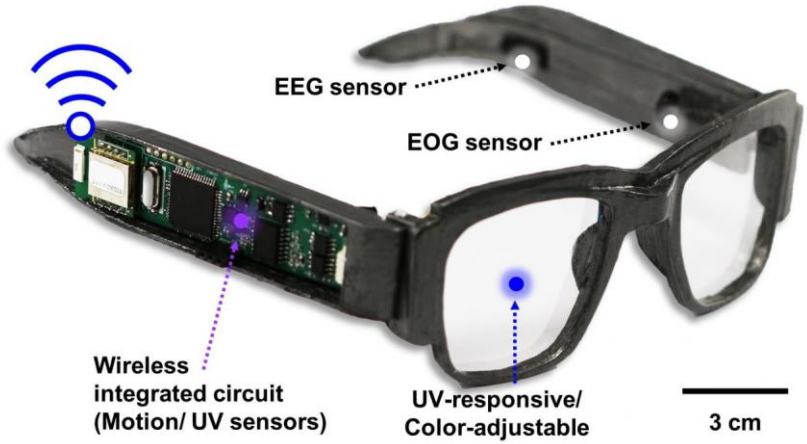
How do you design for the people in the Internet of Things?

- **Design products and services that solve users' pain points and add value to people's lives**
  - UCD places people at the heart of the design process
  - UCD focuses on motivations, goals, behaviors, needs, attitudes, and pain points
- **The key is to understand users**
  - Various techniques (behavioral observation, contextual inquiry, diary studies, surveys, depth interviews, stakeholder workshops, and more…)
- By understanding users' pain points designers can make informed decisions on how to address them

IoT and Ubiquitous Computing

- The IoT deals with how various *things* connected to the internet work <u>together</u>
- Some *things* are portable or wearable
- People use a plethora of different devices and services to support their activities





IoT and Ubiquitous Computing

- Today the interaction with devices is getting more and more **ubiquitous**



- Devices can collect and exploit **"contextual information"**
- **Ubiquitous Computing is computing that occurs anytime and everywhere**

Ubiquitous Computing

- Information processing is more and more integrated into everyday objects
- Computation is finding their way into walls, furniture, as well as objects we carry or wear (smartphones, watches, bracelets, briefcases, purses, wallets, personal digital assistants, etc.)
  - E.g., **ambient intelligence**: sensors in the home environment that can support various functions (e.g., regulating heat, light, music, etc.)
  - E.g., **wearable devices**: sensors collect data about an activity to monitor (e.g., shoes with sensors that understand how you walk)

Ubiquitous Computing



Ubiquitous Interaction

- Typical properties of ubiquitous interaction
  - Occurring all the time
  - Occurring everywhere
  - Transparent to the user
  - Informed by **context**





Context (definitions from Dey & Abowd, 1999)

- *"Context is **any information** that can be used to characterize the situation of an entity. An entity is a person, place, or object that is considered relevant to the interaction between a user and an application, including the user and applications themselves."*

- *"A system is **context-aware** if it uses context to provide relevant information and/or services to the user, where relevancy depends on the user's task."*

Artificial intelligence

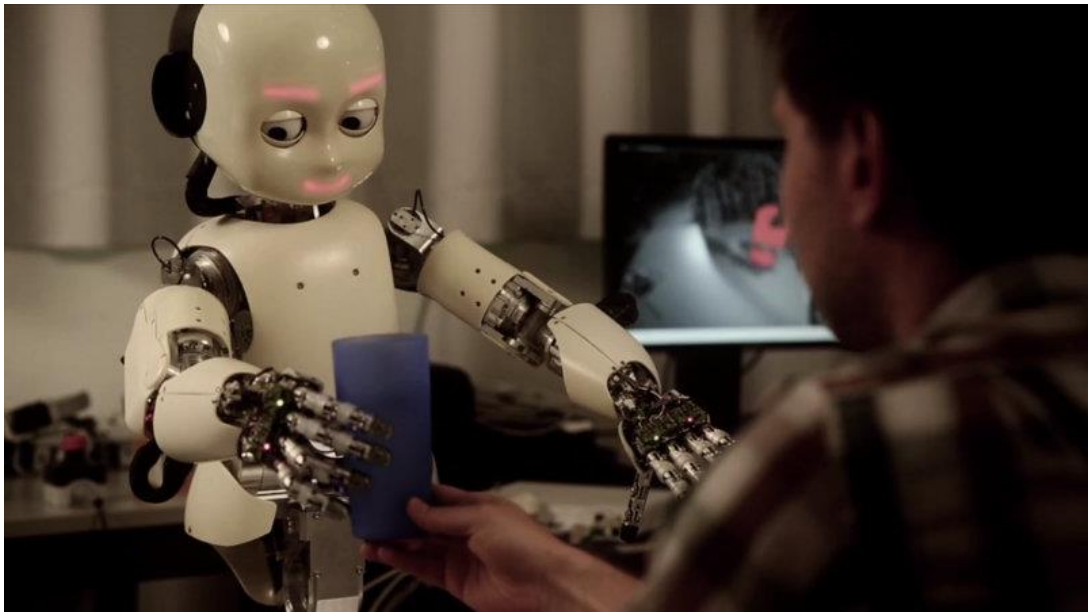
- AI is a set of technologies that allow machines to perceive, understand, act, and learn from experience

- Machine learning techniques are typically used for 3 kind of tasks:
  1. **Classification**
     - E.g., an app that takes as input the images of mushrooms (taken from the smartphone's camera) and classifies mushrooms into 2 categories, edible and poisonous
  2. **Prediction**
     - E.g., an app learns from all the music you listen during the week and predicts which new songs are likely that you like
  3. **Generation**
     - E.g., a user provides a textual prompt about a desired scene and the system generates audio-visual content

HCI and AI

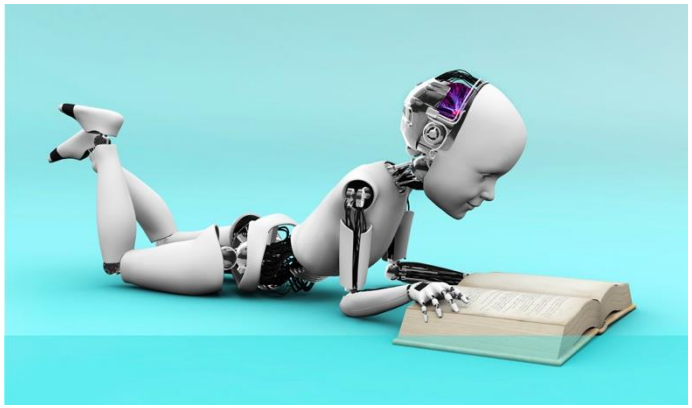
- By developing machine learning techniques to process large volumes and different types of data, and by providing robust predictions of critical outcomes, AI systems can both **supplement** and **replace** human decision-making

- This is rapidly transforming our interactions with machines



HCI and AI

- Objects and environments are increasingly equipped with intelligent features: AI techniques are embedded in these systems to provide **services** to users
- There are **various levels of intelligence** that can be embedded in such systems
- The level of intelligence of a smart object or smart environment **determines the type of interaction** a user will have with it



Intelligence in objects and environments and its relation to interaction

- The interaction with smart objects or smart environments can be categorized as

1. **Strictly tool-based**
2. **Automation**
3. **Proactivity**



Increasing level of intelligence

- The interaction can also be classified as

1. **Explicit**
2. **Implicit**



Increasing level of intelligence

Strictly tool-based

- In this type of interaction users use the smart object or smart environment **consciously**
- Without users no action of the smart object would be triggered; the smart object just supports the human in his actions
- Example: a user wants to switch all the lights in the house off; an application on a smartphone gives the user the opportunity to control the lights



Automation

- Smart object **decides and reacts automatically**. It does not necessarily require the initiation by the user
  - There is not a process of continuous learning from the experience

- Example: the shutters of the building are controlled by sun sensors; if the sun is shining at a specific angle, the shutters are put down

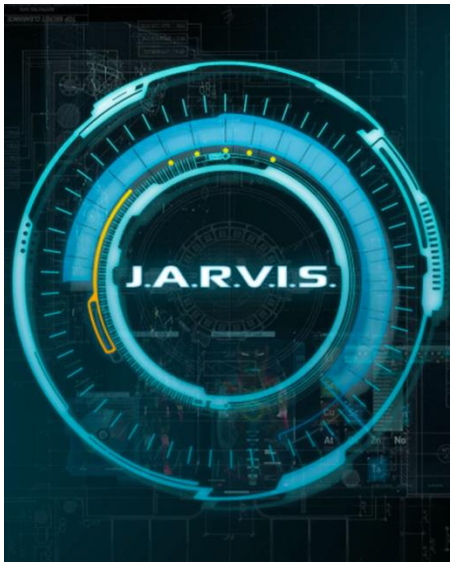


Proactivity

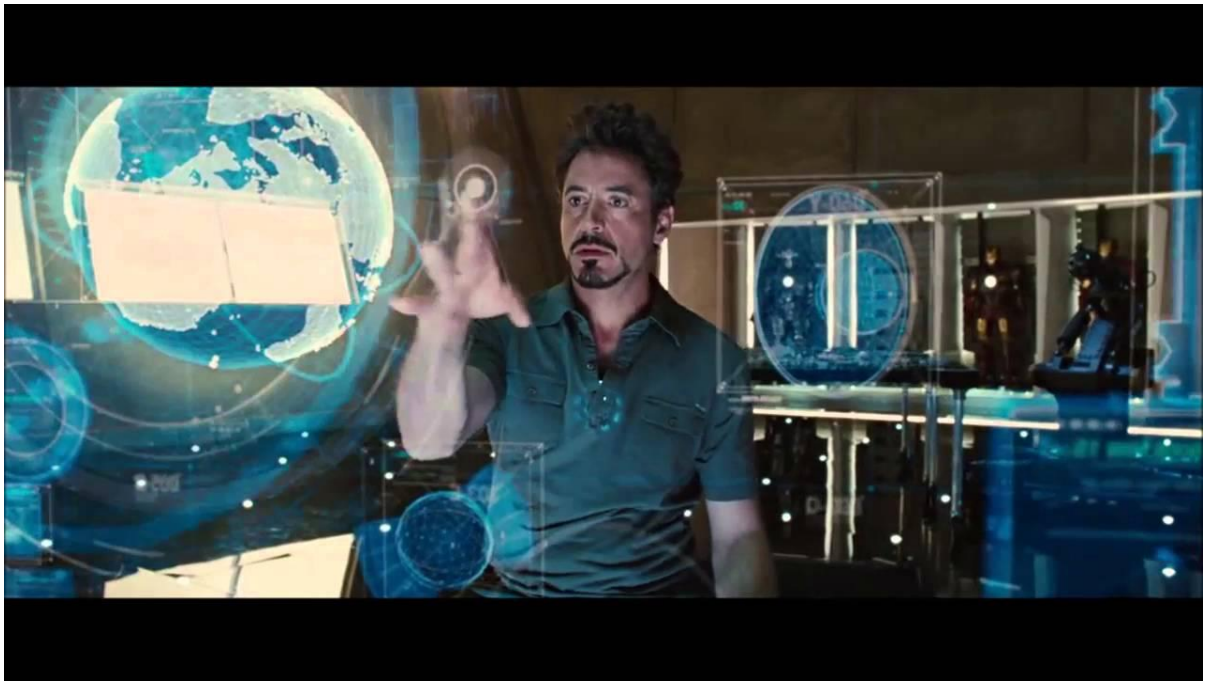
- The smart object/smart environment has the ability to take data from the environment and **make own decisions based on such data**
- The smart object/smart environment is **context-aware** and **learns** from the experience
- Initiation of a functionality is entirely caused by the context, not by the user
- It is the next level of assisted living
- In this interaction model the user puts the most trust in smart objects

Proactivity: example

- The software 'J.A.R.V.I.S' in the movie Iron Man represents the idea of this interaction model; it simulates a human with **self-learning capabilities** and the user interacts and communicates with the software **as if it was also human**







Explicit interaction

- It means that the user is completely **aware** of the smart object s/he uses
- Requires a **direct communication** between the human and the smart object during the whole time of the interaction
- Example: the user uses an app for the smart bulbs in her home to switch the light on and off



Implicit interaction (definitions from Schmidt, 2005)

- **Implicit interaction** is the interaction of a human with the environment and with artefacts which is aimed to accomplish a goal; within this process **the system acquires implicit input from the user and may present implicit output to the user**
- **Implicit input** are actions and behaviour of humans, which are done to achieve a goal and are not primarily regarded as interaction with a computer, but captured, recognized and interpreted by a computer system as input
- **Implicit output** is an output of a computer that is not directly related to an explicit input and which is seamlessly integrated with the environment and the task of the user

Example of implicit interaction

- During the usage the smart toothbrush **receives implicit data from the user**, for example pressure on teeth, and if the pressure is too strong the toothbrush can self-regulate for a healthier usage
- The user does not need a direct communication with the smart object, the communication works in the background

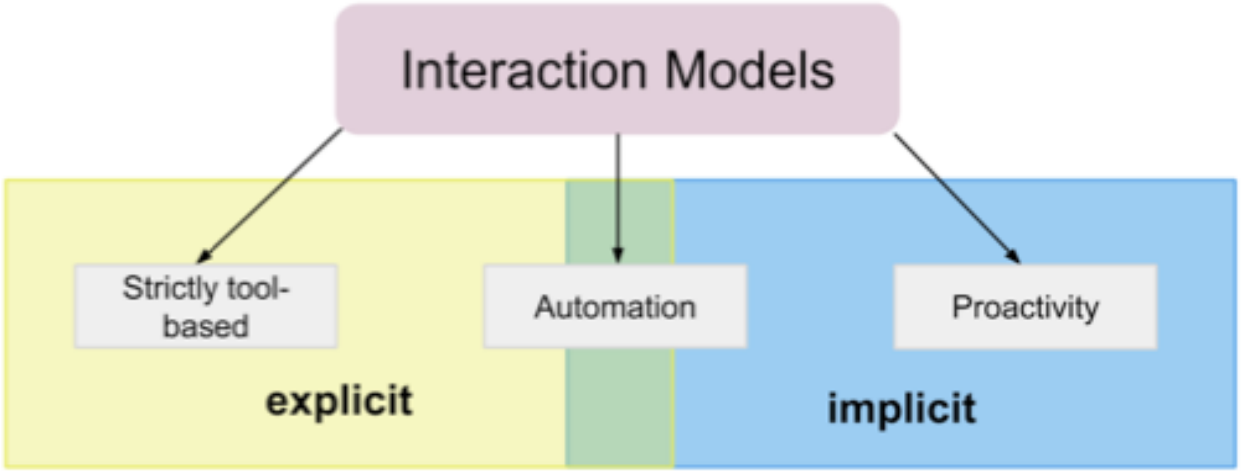


Example of implicit interaction

- The smart fridge registers every product and learns from the owner which product or food she prefers
- When it notices missing food which the owner always had in the fridge, it orders the product on the Internet and informs the owner



Relations between the types of interactions



1. **Strictly tool-based**
2. **Automation**
3. **Proactivity**

Increasing level of intelligence

1. **Explicit**
2. **Implicit**

Increasing level of intelligence

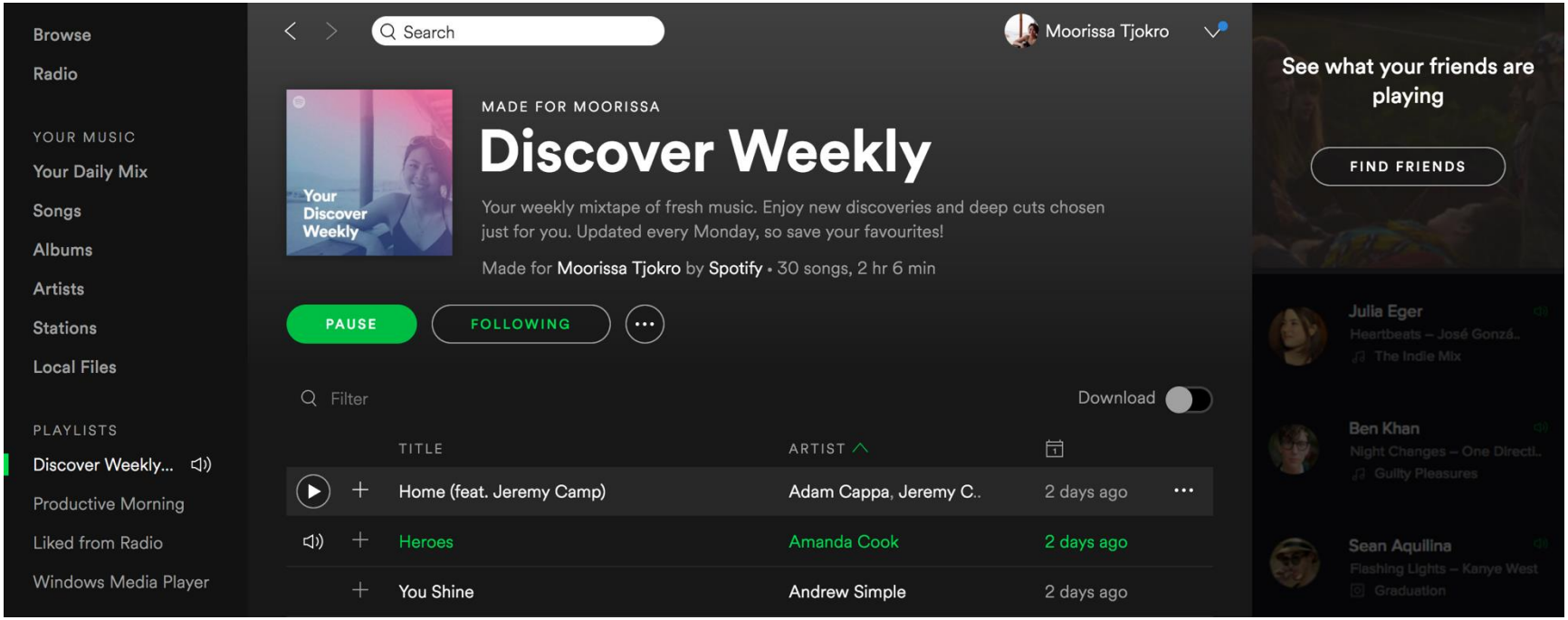
AI systems complexity

- There are various levels of complexity in AI technology
- Simple AI systems: learn from a self-contained dataset and produce a small, fixed set of outputs
  - E.g., face detector system



AI systems complexity

- Complex AI systems: continuously **learn** from the experience and can **adapt** (even in real-time) to the user inputs; they have features such as **context-awareness and proactivity**
  - E.g., music recommender system



Context-awareness

- The use of context is important in interactive applications that use machine learning techniques
- The idea is to exploit the knowledge about the context to provide useful services to users (e.g., recommendations)
- The machines need to learn about the context, i.e., you need to feed them with the **appropriate information as input**
- How do you select this information so that machines take full advantage of the context?
  - **You need to correctly identify the context: this is done by means of user research** (interviews, data logging, observations of users' behaviors, etc.)

Context-awareness

- In order to use context effectively, we must understand both what context is and how it can be used
- We have seen context in the PACT framework
  - **Who** the users are – **People**
  - **What activities** are carried out by users – **Activities**
  - **Where** and **when** the interaction takes place – **Context**
  - **What technologies** are used – **Technologies**

Context-awareness

- In context-aware applications the context refers to all aspects of an activity
- Dey & Abowd's (1999) consider *identity*, *activity*, *time*, and *location* as the primary types of context information, **from which the other contextual information types can be derived**
- Identity, location, and time, answer the fundamental situation-dependent questions of who, where, and when
- Activity answers the question of what is occurring in the situation



This parallels the PACT framework

Context-awareness

- In PACT you need to identify **who** are the users
- There is a branch of research of both AI and HCI that focuses on **user modelling** (or **user profiling**)
- Having this knowledge (e.g., collected from sensors, profiles created online, data logging, etc.) you can tailor a service for that particular user



Proactivity

- Once the context has been identified successfully then it is possible to design services for the users that respond to such context
- **The goal is to improve human decision making**: the proactive application needs to present information in an intuitive and easy to understand format
- Note:

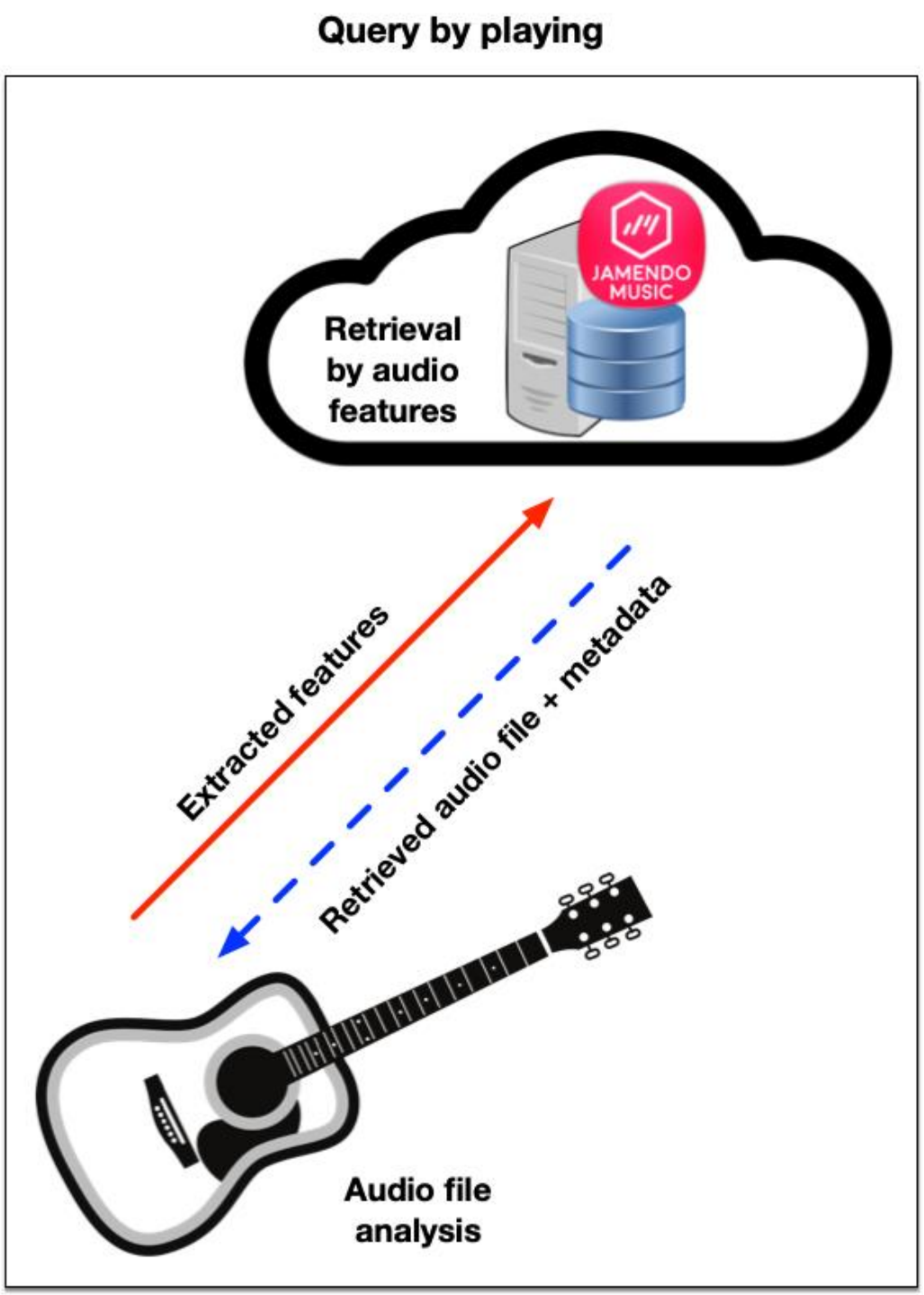
  the context may not be fixed, but **may evolve with time**; therefore, also the proactivity needs to **adapt** to changes of the context

Example: the case of smart musical instruments

- Smart musical instruments are IoT devices enhanced with intelligent capabilities
- Can collect data from the musician via sensors (microphones, knobs, pressure sensors, etc.) and also from data coming from other sources (e.g., social networks)
- The data collected by many instruments can be analyzed to extract patterns (via big data analysis) and then this understanding can be used to better identify in real-time the context around a musician
- From the identified context it is possible to design innovative services for musicians

Example: the case of smart guitar (Turchet et al., 2020)

- Guitarists play over music hosted on online music repositories (e.g., YouTube, Spotify, Jamendo)
- Traditional methods for searching music are text-based (e.g., title of a song, name of an artist)



Example: the case of smart guitar

- Guitarists may want to play over music that contains certain features (e.g., rhythm, mood, chords)
- The intelligence embedded in the instrument understands the properties of a short music piece played by the guitarist with a certain intention (**context-awareness**)
- Then the guitar directly queries to the online repository searching for songs that contain those properties and the repository returns such songs (**proactivity**)
- The guitarist is happy because s/he downloaded a kind of music closer to what s/he needed

How much control to give to users and to AI?

- In designing a product it is important to understand which tasks can be assigned to AI systems and which need to be under the control of the user
- **This decision should be grounded on user research**
- In general, it is important to provide mechanisms of **personalization** of an AI-based system
- In some tasks it is also important to **explain** to users why the AI system has provided a certain recommendation
  - E.g., If a medical-diagnosis intelligent system predicts the likelihood of cancer based on a patient's personal data, the system should provide the reasons for the prediction to the user through a user interface

Laboratory:

work on your exam project

Laboratory

- Get together with your group and start discussing about the possible design of your smart object/smart environment and its interactions
  - Which object/environment do you want to make smart?
    - Are you focusing on an existing object or an entirely new one?
  - Which sensors and actuators? Where are they placed?
  - Which data is collected and which information is provided?
  - Which intelligent capabilities it is equipped with?
  - Which services does it offer?
  - Which types of interactions it can afford?

Project: questions when designing your smart object/environment and its interactions

- What are the types of interactions for your project?
  - Strictly tool-based, automation, or proactivity?
  - Explicit or implicit, or both?
  - This means, **how much control you want to give to users and to the intelligence embedded in the system?**
- Note: these interactions are combined with those seen in previous lesson
  - Discrete, continuous, touch-based, touchless, synchronous, asynchronous, individual, collaborative, situated, remote, networked, unimodal or multimodal

Project: smart objects and smart environments

- You can choose to focus your project on the smart objects or smart environments you prefer, e.g.:
  - Home appliances
  - Personal devices
  - Wearable devices
  - Devices for smart agriculture
  - Devices for traffic management
  - Devices for manufacturing in Industry 4.0
  - ….

Project: assumptions you can make

- Without doing an implementation, **you can assume that**:
  - Your system is structured around a network of devices
  - Your system is able to fuse information from different sources
  - There may be algorithms for classification and inference on big data collected by several devices (e.g., pattern recognition, speech recognition, translation, face recognition, object recognition, etc.)
- To the test the system you can use evaluation techniques that do not require the implementation (e.g., we will see the "Wizard of Oz" technique)

Project: considerations you should make

- Consider:
  - Do you want to create a context-aware recommender system that is embedded into a physical everyday object? Or rather an app or website?
  - What is the context for your intelligent application?
  - How do you acquire the context?
  - What are the most important data to collect from sensors, extract, and display?
  - Why the information displayed are relevant to users?
  - How complex is the AI system (e.g., simple classification vs. continuously learning and adaptation)?
  - What is the degree of autonomous behavior the AI system has and how much control the user has during each task?

THIS YEAR PROJECTS FOCUS

- **This year projects must integrate at least 3 among these aspects:**
  - Use of the Internet of Things paradigm
  - AI methods for classification / prediction / generation
  - Gestures tracking
  - Biometric sensing (e.g., Brain computer interfaces, heart rate)
  - Environmental quantities sensing
  - Sensor fusion
  - Multisensory input/output
- This means that a **strong** focus on some of these aspects needs to be placed in the ideation phase of the solution
- However, the solution needs to be designed starting from the actual **user requirements**

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